

Name (Pinyin): _____

姓名: _____

S&T2025 Basic C Programming Language Mock Test (First Checkpoint)

Time allowed: 1 hour and 30 minutes

INSTRUCTIONS

1. This is a **CLOSED BOOK** examination.
2. This examination paper contains **10 questions** and 12 pages. Answer **all** questions.
3. The full score for each question is as follows:

Question 1: 5	Question 4: 5	Question 7: 10	Question 10: 30
Question 2: 5	Question 5: 5	Question 8: 10	
Question 3: 5	Question 6: 5	Question 9: 20	
4. Answers to the questions are to be written in this booklet.
5. Students are **NOT** allowed to bring any programmable calculator, machine translator, or dictionary with them. If you have any of these items, you are to surrender them before the test starts. **Violation of this regulation will be severely dealt with and may result in an expulsion from the bridging course.**
6. Students are also **NOT** allowed to bring any paper to the Examination Hall.
7. All the working steps must be shown.
8. This test paper will be collected at the end of the examination.

No.	Score 1 (Marker)	Score 2 (Checker)
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

1. What will be printed on the screen after the following C program is executed?

```
#include <stdio.h>

int main()
{
    int i = 1;
    switch (i==0)
    {
        case 0: printf ("i=0");
                break;
        case 1: printf ("i=1");
                break;
        default: printf ("i=20");
                break;
    }
    return 0;
}
```

(5 Marks)

Answer:

2. What will be printed on the screen after the following C program is executed?

```
#include <stdio.h>

void main()
{
    int x=19, y=47;
    x= y++;
    y= ++y + ++x;
    printf("x=%d y=%d",x,y);
}
```

(5 Marks)

Answer:

3. What is the numerical value of the following expression in C language?

!(12>11)

(5 Marks)

Answer:

4. How many times will the following for loop execute?

```
# include <stdio.h>

void main ()
{
    int i, j;
    for (i=10;j=1;j++) printf("==>%d",j);
}
```

(5 Marks)

Answer:

5. What will be printed on the screen after the following C program is executed?

```
# include <stdio.h>

int a=0;

void change (int a)
{
    a=a-1;
}

void main()
{
    int a=10;

    change (a);

    a=a-1;
    printf("a=%d",a);
}
```

(5 Marks)

Answer:

6. How many times will the following do loop execute?

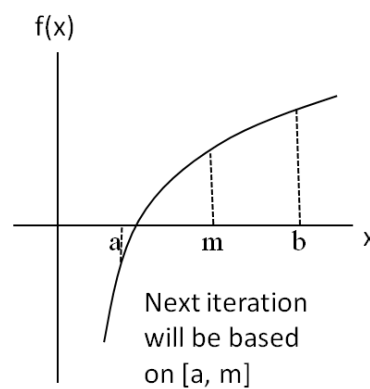
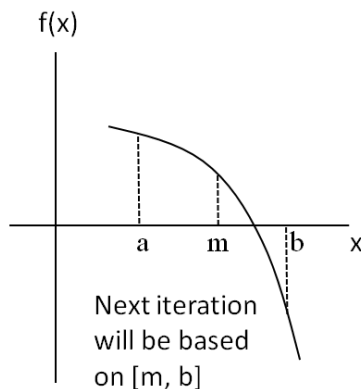
```
# include <stdio.h>
void main()
{
    int k=0;

    do
    {
        k++;
        printf("%d ",k);
        k=2*k;
    }
    while(k<=12);
}
```

(5 Marks)

Answer:

7. Bisection method is to divide the interval of a function into two equal parts. The method can be used to approximate the root of function. A simple bisection procedure for iteratively converging to a solution which is known to lie inside some interval $[a, b]$, where $f(a) \times f(b)$ will have to be negative, proceeds by evaluating the function in question at the midpoint m , $m = (a+b)/2$, from the original interval $[a, b]$. If $f(a)$ and $f(m)$ are of the same sign, the iteration will proceed by replacing $[a, b]$ by $[m, b]$ if needed (refer to Figure (i)). Similarly, if $f(b)$ and $f(m)$ are of the same sign, the bisection method will use $[a, m]$ in its next iteration if needed (refer to Figure (ii)). As the intervals used in the subsequent iterations are becoming smaller and smaller due to the halving operation, the bisection method can approximate the root of the function when $f(m)$ is within the accepted tolerance.

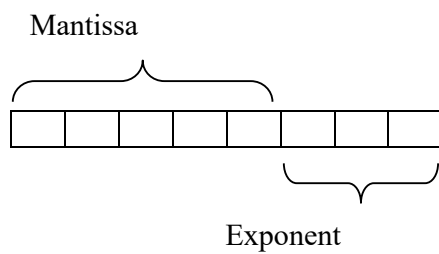


Write a C program based on the bisection method to approximate the value of $\sqrt[3]{279}$. You can use $[1, 3]$ as the starting interval.

(10 Marks)

Answer:

8. Represent $-0.4375_{(10)}$ by a floating-point notation with 5 mantissa bits and 3 exponent bits as follows.



Answer:

(10 Marks)

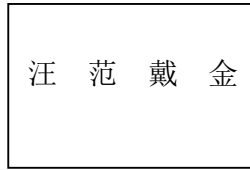
9. A stick of 1 m is cut at random into two pieces. Write a C program which uses **random numbers** to approximate the following two probabilities.
- (i) What is the probability that the product of the lengths of the pieces is at most $\frac{3}{16}m^2$? Print the approximation on the screen.
 - (ii) What is the probability that the absolute difference of the lengths of the pieces is not larger than $\frac{9}{16}m$. Print the approximation on the screen.

Mathematical solution is not required and will not be given any marks.

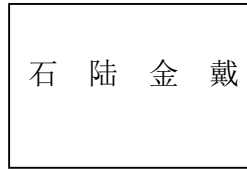
(20 Marks)

Answer:

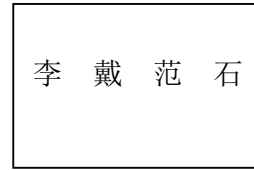
10. Many fortune tellers play this game to garner the confidence of their potential customers. The customers are given cards containing different numbers of family names. The customers then pick up all the cards that contain his/her family name. For example, if your family name is 金, and you are given the following cards:



Card-I



Card-II



Card-III

You will pick up Card-I and Card-II for the fortune teller. Although six family names: 汪, 范, 戴, 金, 石, 陆 appear on the two cards, the fortune teller will confidently say that your family name is 金, and claims that he/she has the supernatural ability to tell your future. But actually, this game has nothing to do with the supernatural ability.

Use 7 distinct family names (汪, 陆, 金, 李, 范, 石, 戴) as an example and the binary representation to answer the following questions.

- (i) Explain how the cards are designed to guess the 7 names. How many cards are necessary and sufficient to contain the 7 distinct names in order to have the correct guess at all times?
- (ii) How many cards are necessary and sufficient to have correct guess from the 100 distinct family names in the “百家姓”?
- (iii) Explain why your mechanism is able to guarantee the correct guess each time the game is played, if the family name of the customer is in the cards.

(30 Marks)

Answer:

